CSCI E-106: TA Section

March 3 2026

Contents


```
## ggplot2 loaded properly
## knitr loaded properly
## MASS loaded properly
## formatR loaded properly
```

(Textbook 3.3) Refer to Grade point average Problem 1.19.

Please use dataset titled: CH01PR19.txt**

a. Load your data & create regression model

```
Dataset_1_19 = read.table("CH01PR19.txt", header = FALSE, sep = "", col.names = c("V1",
    "V2"))

lmfit19 = lm(V1 ~ V2, data = Dataset_1_19)


summary(lmfit19)
```

```
##
## Call:
## lm(formula = V1 ~ V2, data = Dataset_1_19)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.74004 -0.33827  0.04062  0.44064  1.22737
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.11405    0.32089   6.588  1.3e-09 ***
## V2           0.03883    0.01277   3.040  0.00292 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6231 on 118 degrees of freedom
## Multiple R-squared:  0.07262,    Adjusted R-squared:  0.06476
## F-statistic:  9.24 on 1 and 118 DF,  p-value: 0.002917
```

```
# save our residuals
ei = lmfit19$residuals

# rank our residuals from small to large
```

```r
ri = rank(ei)
```

e. Conduct the Brown-Forsythe test to determine whether or not the error variance varies with the level of X. Divide the data into the two groups, X < 26, X >= 26, and use a = .01. State the decision rule and conclusion. Does your conclusion support your preliminary findings in part (c)?

```r
summary(lmfit19)
```

```
##
## Call:
## lm(formula = V1 ~ V2, data = Dataset_1_19)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.74004 -0.33827  0.04062  0.44064  1.22737
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## Coefficients:
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```

```r
# Remember what brown-forsythe is for - checking error variance is equal across
# all

#---this is the long hand way---

# break into two groups
df = data.frame(cbind(Dataset_1_19[, 1], Dataset_1_19[, 2], ei))
df1 = df[df[, 2] < 26, ]
df2 = df[df[, 2] >= 26, ]

summary(df1[, 3])
```

```
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
## -1.243728 -0.390900 -0.032900  0.005155  0.427581  1.227371
```

```r
summary(df2[, 3])
```

```
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
## -2.740036 -0.262709  0.142618 -0.006092  0.520464  0.798791
```

```r
# n1
n1 = length(df1[, 3])
print(n1)
```

```
## [1] 65
```

```r
# n2
n2 = length(df2[, 3])
print(n2)
```

```
## [1] 55
```

```r
# this is the long hand for taking our absolute values of the residuals - the
# median of the selected group find the median from above
d1 = abs(df1[, 3] + 0.0329)
d2 = abs(df2[, 3] - 0.142618)

# calculate means for our answer
mean(d1)
```

```
## [1] 0.4379603
```

```r
mean(d2)
```

```
## [1] 0.5065161
```

```r
s2 = (var(d1) * (65 - 1) + var(d2) * (55 - 1))/(120 - 2)
print(s2)
```

```
## [1] 0.1741184
```

```r
# calculate s
s = sqrt(s2)
print(s)
```

```
## [1] 0.4172749
```

```r
# manually creating t statistic testStastic = (mean.d1 - mean.d2) / (s *
# sqrt((1/n1)+1/n2)
testStastic = (0.43796 - 0.50652)/(0.417275 * sqrt((1/65) + (1/55)))
print(testStastic)
```

```
## [1] -0.8968005
```

```r
t = qt(0.995, 118)
print(t)
```

```
## [1] 2.618137
```

Part E Notes: We need to put our answer together

n1 = 65, mean.d1 = .43796 n2 = 55, mean.d2 =.50652 s = .417275 test Statistic = (.43796 - .50652)/ .417275 q(1/65)+(1/55) = -.8968005 t =(.995,118) = 2.61814

Part E Conclusion: If abs(testStatistic) <= 2.61814 conclude error variance constant, otherwise error variance not constant. Thus, we conclude error variance constant.

```r
#---this is a shorter way---
# break into groups:
group <- ifelse(Dataset_1_19$V2 < 26, "X<26", "X>=26")

# group medians
med1 <- median(ei[group == "X<26"])
med2 <- median(ei[group == "X>=26"])

# absolute deviations
z <- ifelse(group == "X<26", abs(ei - med1), abs(ei - med2))

# one-way ANOVA of our Brown-Forsythe test)
bf <- aov(z ~ group)
summary(bf)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## group         1   0.14  0.1400   0.804  0.372
## Residuals   118  20.55  0.1741
```

decision rule: if p <= .01, reject H0 (variances are different), if p > .01 fail to reject H0 (no variance differing)

so we fail to reject since our p value is: 0.372

(Textbook 3.32)

Refer to the Prostate cancer data set in Appendix C.5. Build a regression model to predict PSA level (Y) as a function of cancer, Volume (X). The analysis should include an assessment of the degree to which the key regression assumptions are satisfied. If the regression assumptions are not met, include and justify appropriate remedial measures. Use the final model to estimate mean PSA level for a patient whose cancer volume is 20 cc. Assess the strengths and weaknesses of the final model.

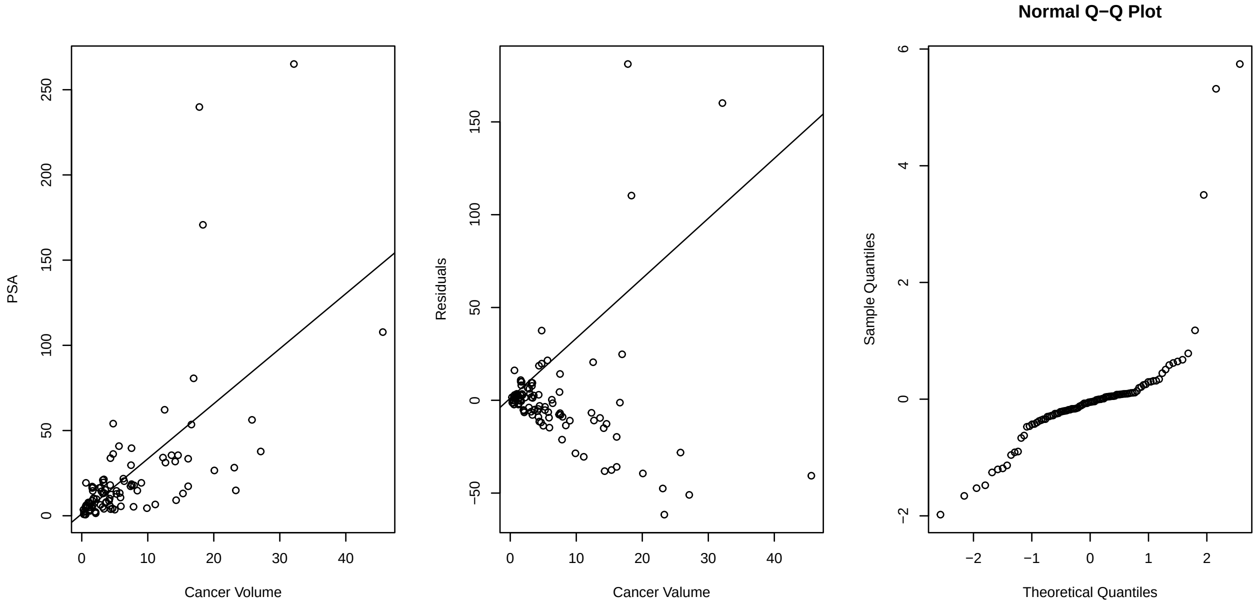
Please use dataset titled: ***APPENC05.txt***

```r
df_prostate = read.table("APPENC05.txt", header = FALSE, sep = "", col.names = c("id",
    "psa", "cancerVol", "weight", "age", "benPros", "seminalVes", "capPen", "gleasonScore"))


f_3.32 = lm(df_prostate$psa ~ df_prostate$cancerVol)
summary(f_3.32)
```

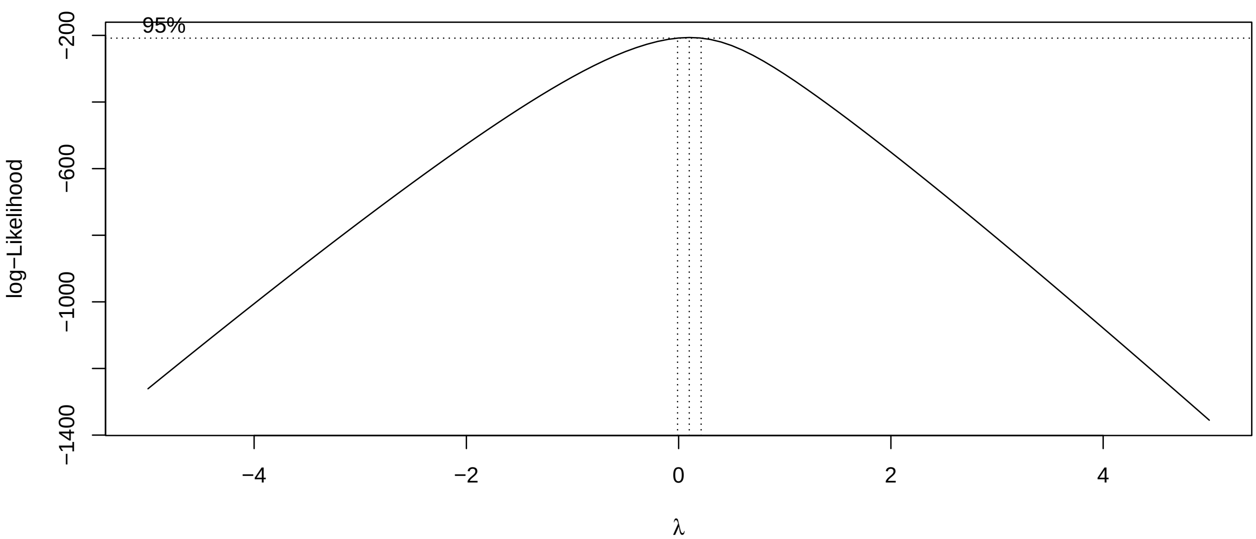
```
##
## Call:
## lm(formula = df_prostate$psa ~ df_prostate$cancerVol)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -61.619  -9.023  -1.586   3.151 181.183
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)             1.1249     4.3596   0.258    0.797
## df_prostate$cancerVol   3.2299     0.4148   7.786 8.47e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 32.03 on 95 degrees of freedom
## Multiple R-squared:  0.3896, Adjusted R-squared:  0.3831
## F-statistic: 60.63 on 1 and 95 DF,  p-value: 8.468e-12
```

```r
e1 = f_3.32$residuals

# standardize residuals needed for QQ plot##
rs1 = rstandard(f_3.32)

par(mfrow = c(1, 3))

plot(df_prostate$cancerVol, df_prostate$psa, xlab = "Cancer Volume", ylab = "PSA")
abline(f_3.32)

plot(df_prostate$cancerVol, e1, xlab = "Cancer Valume", ylab = "Residuals")
abline(f_3.32)
```

```r
## QQ plot ##
qqnorm(rs1)
```




```r
par(mfrow = c(1, 1))

boxcox(f_3.32, lambda = seq(-5, 5, by = 0.1))
```

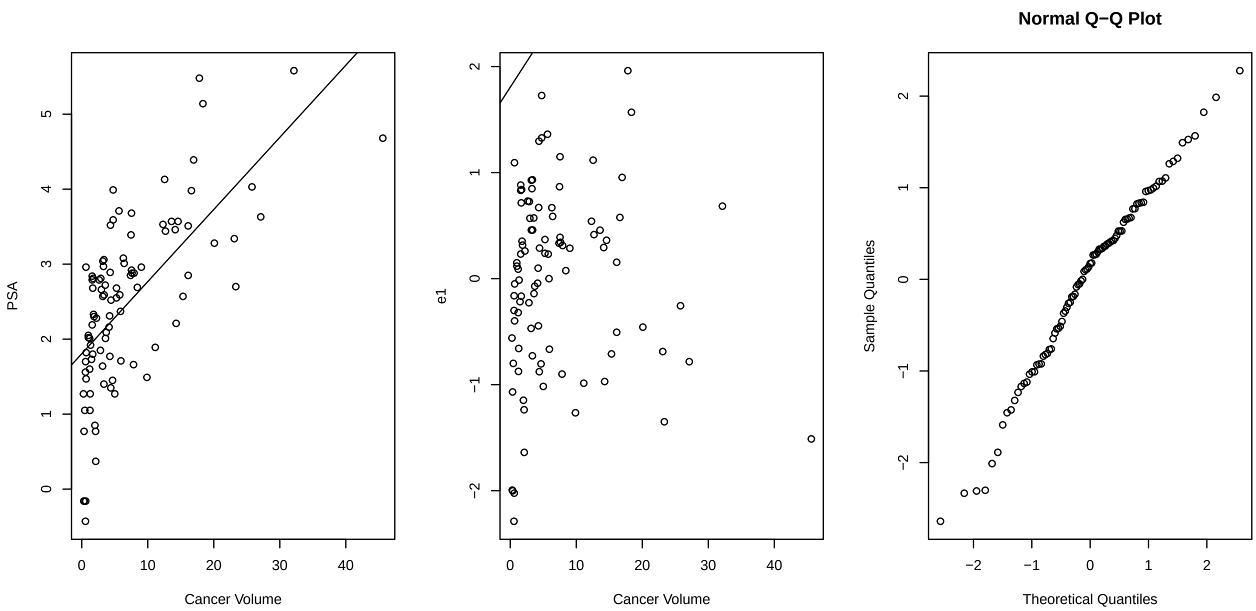



```r
# From our boxcox we see that we need to use the log transformation

f_3.32_2 = lm(log(df_prostate$psa) ~ df_prostate$cancerVol)
summary(f_3.32_2)
```

```
##
## Call:
## lm(formula = log(df_prostate$psa) ~ df_prostate$cancerVol)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
```

```
## -2.2886 -0.6590  0.1493  0.5769  1.9610
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)            1.80549    0.11899  15.174  < 2e-16 ***
## df_prostate$cancerVol  0.09619    0.01132   8.496 2.69e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8742 on 95 degrees of freedom
## Multiple R-squared:  0.4317, Adjusted R-squared:  0.4258
## F-statistic: 72.18 on 1 and 95 DF,  p-value: 2.688e-13
```

```r
e1 = f_3.32_2$residuals

# standardize residuals needed for QQ plot##
rs1 = rstandard(f_3.32_2)

par(mfrow = c(1, 3))

plot(df_prostate$cancerVol, log(df_prostate$psa), xlab = "Cancer Volume", ylab = "PSA")

abline(f_3.32_2)

plot(df_prostate$cancerVol, e1, xlab = "Cancer Volume")

abline(f_3.32_2)

qqnorm(rs1)
```



Problem 3.32: It still seems like we were getting a lot of outliers in our 2 graphs. Our normal plot is in line but not still not completely a great line.